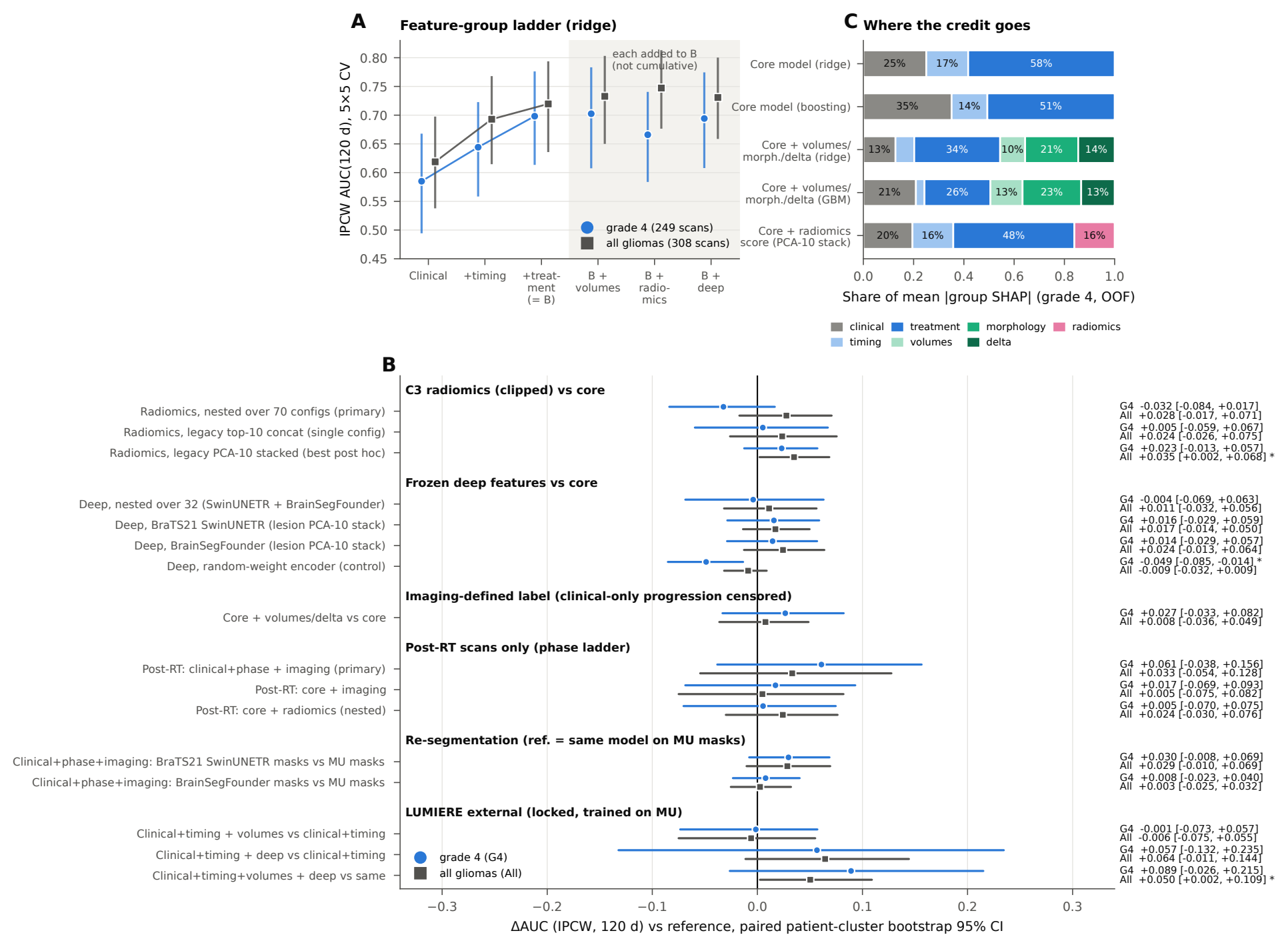
A
Feature-group ladder (ridge)
IPCW AUC(120 d), 5×5 CV
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
each added to B
(not cumulative)
grade 4 (249 scans)
all gliomas (308 scans)
Clinical
+timing
+treat-
ment
(= B)
B +
volumes
B +
radio-
mics
B +
deep
C Where the credit goes
Core model (ridge)
25%
17%
58%
Core model (boosting)
35%
14%
51%
Core + volumes/
morph./delta (ridge)
13%
34%
10%
21%
14%
Core + volumes/
morph./delta (GBM)
21%
26%
13%
23%
13%
Core + radiomics
score (PCA-10 stack)
20%
16%
48%
16%
0.0
0.2
0.4
0.6
0.8
1.0
Share of mean |group SHAP| (grade 4, OOF)
clinical
treatment
morphology
radiomics
timing
volumes
delta
B
C3 radiomics (clipped) vs core
Radiomics, nested over 70 configs (primary)
G4 -0.032 [-0.084, +0.017]
All +0.028 [-0.017, +0.071]
Radiomics, legacy top-10 concat (single config)
G4 +0.005 [-0.059, +0.067]
All +0.024 [-0.026, +0.075]
Radiomics, legacy PCA-10 stacked (best post hoc)
G4 +0.023 [-0.013, +0.057]
All +0.035 [+0.002, +0.068] *
Frozen deep features vs core
Deep, nested over 32 (SwinUNETR + BrainSegFounder)
G4 -0.004 [-0.069, +0.063]
All +0.011 [-0.032, +0.056]
Deep, BraTS21 SwinUNETR (lesion PCA-10 stack)
G4 +0.016 [-0.029, +0.059]
All +0.017 [-0.014, +0.050]
Deep, BrainSegFounder (lesion PCA-10 stack)
G4 +0.014 [-0.029, +0.057]
All +0.024 [-0.013, +0.064]
Deep, random-weight encoder (control)
G4 -0.049 [-0.085, -0.014] *
All -0.009 [-0.032, +0.009]
Imaging-defined label (clinical-only progression censored)
Core + volumes/delta vs core
G4 +0.027 [-0.033, +0.082]
All +0.008 [-0.036, +0.049]
Post-RT scans only (phase ladder)
Post-RT: clinical+phase + imaging (primary)
G4 +0.061 [-0.038, +0.156]
All +0.033 [-0.054, +0.128]
Post-RT: core + imaging
G4 +0.017 [-0.069, +0.093]
All +0.005 [-0.075, +0.082]
Post-RT: core + radiomics (nested)
G4 +0.005 [-0.070, +0.075]
All +0.024 [-0.030, +0.076]
Re-segmentation (ref. = same model on MU masks)
Clinical+phase+imaging: BraTS21 SwinUNETR masks vs MU masks
G4 +0.030 [-0.008, +0.069]
All +0.029 [-0.010, +0.069]
Clinical+phase+imaging: BrainSegFounder masks vs MU masks
G4 +0.008 [-0.023, +0.040]
All +0.003 [-0.025, +0.032]
LUMIERE external (locked, trained on MU)
Clinical+timing + volumes vs clinical+timing
G4 -0.001 [-0.073, +0.057]
All -0.006 [-0.075, +0.055]
Clinical+timing + deep vs clinical+timing
G4 +0.057 [-0.132, +0.235]
All +0.064 [-0.011, +0.144]
Clinical+timing+volumes + deep vs same
G4 +0.089 [-0.026, +0.215]
All +0.050 [+0.002, +0.109] *
grade 4 (G4)
all gliomas (All)
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
ΔAUC (IPCW, 120 d) vs reference, paired patient-cluster bootstrap 95% CI